AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM112

Introduction to Computing and Information Technology

Final Examination
Fall Semester 2024/2025

Date: 15/Jan./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 3 hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EGYPT

CamScanner

Part I: Multiple Choice Questions (20 marks).

1- Possible use cases of where the Internet of Things is expected to be used:
   a- Utilities (gas, water, electricity)
   b- Smart cities
   c- Healthcare
   d- All of the above

2- Which of the following is an example of Infrastructure as a Service, IaaS?
   a. Providing customers with a web-based e-mail service
   b. Providing an environment in which the customers' applications can be run
   c. Providing customers with access to virtual machines.
   d. None of the given choices.

3- Collision happens when:
   a. identical hashes for different pieces of data
   b. different hashes for different pieces of data
   c. different hashes for identical pieces of data
   d. None of the given.

4- This is a piece of software that enables the operating system to read and write data to a disk image
   a. image mounter
   b. Metadata
   c. Timestamps
   d. write blocker

5- ..... is a type of malware that prevents a user from accessing their computer either by locking the computer screen or by blocking access to computer files until a 'ransom' is paid.
   a. Ransomware
   b. Malware
   c. Brute force
   d. Encryption

6- Interference..... the probability of error; and the stronger the interference, the ..... the probability of error.
   a. Decreases, smaller
   b. Decreases, greater
   c. Increases, smaller
   d. Increases, greater

7- If we use 5-bit binary numbers, what is the maximum wind speed that can be represented?
   a- 4
   b- 5
   c- 31
   d- 32

8- Encryption has the security benefit of
   a- confidentiality
   b- integrity
   c- authentication
   d- all the above

9- ---------- a malicious user secretly interposed between sender and receiver, which modifies or deletes information
   a. Integrity
   b. Man-in-the middle attack
   c. Authentication
   d. Confidentiality

10- Denial-of-service (DoS)
   a- produced through a complex mathematical process requiring huge amounts of computer power
   b- method of attacking computers by flooding their network connections with spurious data that prevents legitimate traffic from being sent or received
   c- Passwords and other forms of secure data can be attacked by brute force
   d- used to send spam messages to every contact in their address book.

YOU EGYPT

CamScanner

Part 2: Short questions (30 marks)

Question 1: (6 marks)

What are the types of real time operating system and give the definition of each of them?

Question 2: (6 marks)

a) What fraction of the power would be received by the antenna at d = 4?
b) Show that the binary number 1111.111 represents the denary number 15.875.

Question 3: (6 marks)

Define three user issues in the cloud

Question 4: [6 marks]
Give the definition of TLS/SSL and mention two services that TLS/SSL provides?

Question 5: [6 marks]

a) How does the operating system know where to find the file again once a file has been written to one or more clusters of a hard disk?
b) What happens when a file is deleted from a hard disk?

Part 3: Problem Solving / Long Questions (50 marks)

Question 1: (10 marks)

a- What types of physical media might be used to transmit the data?
b- The following statements apply to either TCP or IP. Select the correct option in each case (either IP or TCP).
1. It sets up connections in order to send data between two computers.
2. It operates in the internet layer.
3. It addresses packets and routes them across networks.
4. It operates in the transport layer.
5. It deals with missing packets.

Question 2: (10 marks)

Mention and describe the three crucial properties of Hashing?

Question 3: [10 marks]
How are Internet messages delivered?

Question 4: [10 marks]

a. What is the different between **circuit-switched and packet switching? (6 marks)**
b. What are the Technical features that indicate why the internet has become successful?

Question 5: [10 marks]

Give two common techniques use for attacking passwords and give example on each

---

End of Questions

AOU EGYPT

CamScanner